

RMO(INDIA) – 1986(1)

Max Mark – 100

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
- Ruler and compass allowed.
- Answer all the questions.

1. Prove that the fraction $\frac{21n+4}{14n+3}$ is reducible for every natural number n .
2. Let a, b, c be real numbers, consider the quadratic equation $a \cos^2 x + b \cos x + c = 0$. Using the numbers a, b, c form a quadratic equation in $\cos 2x$ whose roots are the same as those of the original equation. Compare the equations in $\cos x$ and $\cos 2x$ for $a = 4, b = 2$ and $c = 1$.
3. Determine all three digit numbers N , having the property that N is divisible by 11 and $N/11$ is equal to the sum of the squares of the digits of N .
4. For what real values of the variable x does the following inequality hold $\frac{4x^2}{(1-\sqrt{1+2x})^2} < 2x+9$.
5. Consider an isosceles triangle. Let r be the radius of its circumscribed circle and p the radius of the inscribed circle. Prove that the distance d between these two centres is $d = \sqrt{r(r-2p)}$.
6. Consider a triangle $P_1P_2P_3$ and a point P within the triangle. Lines P_1P, P_2P, P_3P intersect the opposite sides in points Q_1, Q_2 and Q_3 respectively. Prove that out of the numbers $\frac{P_1P}{PQ_1}, \frac{P_2P}{PQ_2}, \frac{P_3P}{PQ_3}$ atleast one is ≤ 2 and atleast one is ≥ 2 .
7. Find the smallest natural number n which has the following properties:
 - (i) Its decimal representation has 6 as the last digit.
 - (j) If the last digit is erased and placed in front of the remaining digits, the resulting number is four times as large as the original number n .
8. When 4444^{4444} is written in decimal notation the sum of its digit is A . Let B be sthe sum of the digits of A . Find the sum of the digits of B .
9. Solve the equation: $\cos^2 x + \cos^2 2x + \cos^2 3x = 1$

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